Special Boundary Roles in the Innovation Process

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The research described in this article focuses on one important aspect of the innovation process — the need for the innovating system to gather information from and transmit information to several external information areas. Special boundary roles evolve in the organization's communication network to fulfill the essential function of linking the organization's internal network to external sources of information. These boundary roles occur at several organizational boundaries, and their distribution within the organization is contingent on the nature of the organization's work. This research supports literature on boundary spanning in general and highlights the importance of boundary roles in the process of innovation.•

Social scientists have long been interested in the development and the diffusion of innovations (Myers and Marquis, 1969; Rogers and Shoemaker, 1971; Achilladeles, Jervis, and Robertson, 1971; Baldridge and Burnham, 1975). This interest in innovation has extended over a range of organizations (for example, research and development laboratories, health and welfare organizations, and school systems) as well as over a range of innovations (for example, hybrid corn, drugs, types of social service, and new technologies).

One critical aspect of the innovation process is the ability of the innovating unit to gather information from and transmit information to several external domains (Myers and Marquis, 1969; Miller, 1971; Kelly and Kranzberg, 1975). Yet, while information from external areas is crucial to the effective development or diffusion of innovation, substantial literature suggests that communication across organizational boundaries is both inefficient and prone to bias and distortion (Dearborn and Simon, 1958; March and Simon, 1958; Wilensky, 1967). There is, then, a dilemma — accurate information from external areas is vital to the innovation process yet relatively difficult to gather. This dilemma and its consequences are an important impediment to the innovation process (Utterback, 1974; Mansfield and Wagner, 1975).

Building on the innovation-oriented research of the organizational behavior and R&D management literature, this study investigates the existence and the characteristics of special boundary roles as one means for innovating organizations to deal with the necessity of cross-boundary communication. The distribution of these special boundary roles within the organization and their impact on subunit performance is also investigated. While the setting for this field study was an R&D laboratory,[1] the thrust and logic of this paper are directed toward the more generally interesting question of the place of boundary roles in the innovation process.

INNOVATION AND THE IMPORTANCE OF COMMUNICATION ACROSS SEVERAL BOUNDARIES

Zaltman, Duncan, and Holbek (1973) have argued that the development and diffusion of innovations can be seen as a process composed of several phases. Each phase is charac-

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1 This study was conducted at the R&D facility of a major American corporation. The facility is divided into seven departments, which are, in turn, divided into projects. Projects range from basic research to technical service. Details appear in the methods section.

terized by different types of decisions and problems of coordination as well as by different patterns of communication (Zaltman, Duncan, and Holbek, 1973; Allen, 1977). Consistent with these ideas, Myers and Marquis (1969) have proposed a three-step innovation process: idea development (the generation of a design concept), problem solving (technical efforts and problem solving in developing the proposed idea), and implementation (pilot production, interarea coordination). Baker, Siegmann, and Rubenstein (1967) and Mansfield *et al.* (1971) have suggested similar processes based on R&D settings; Van de Ven and Koenig (1976) have developed a similar approach based on innovation in social service settings.

While this process approach to innovation suggests a range of problem solving differences in the various phases, one important difference is the locus of critical information and feedback (Figure 1). During the first phase, idea generation,

Innovation Phases		
I. Idea Generation: Developing a design or proposal by integrating market need with technical means.	II. Problem Solving: Research, development, and engineering to develop a technical solution to the proposal.	III. Implementation: R&D, manufacturing, market tooling up, coordination, and administration to bring the new product to first use or to the market.
Key Communication Domains by Phase		
I. Extra-laboratory, extra-organizational communication: With universities, suppliers, vendors, literature.	II. Intra-laboratory communication: With functional areas as well as technical areas outside laboratory.	III. Laboratory and functional area communication: Particularly between R&D and manufacturing and marketing

Figure 1. Innovation phases and corresponding key communication domains. (Figure adapted from Myers and Marquis [1969]).

information pertaining to new ideas or new approaches to technical or marketing problems can be found most effectively outside the innovating organization (Utterback, 1971). In the second phase, the more focused problem solving requires exchanging substantial information within the R&D subunit as well as between the laboratory and its manufacturing and marketing areas (Pelz and Andrews, 1966; Gerstenfeld, 1970). The third phase, implementation, demands effective coordination and problem solving among all the functional areas (Morton, 1971; Mansfield and Wagner, 1975).

This process-oriented approach to innovation emphasizes the importance of several external sources of information (from the point of view of the innovating unit). While need for communication and information will vary over time, this approach suggests that to be effective, the innovating subunit must efficiently gather information from and transmit

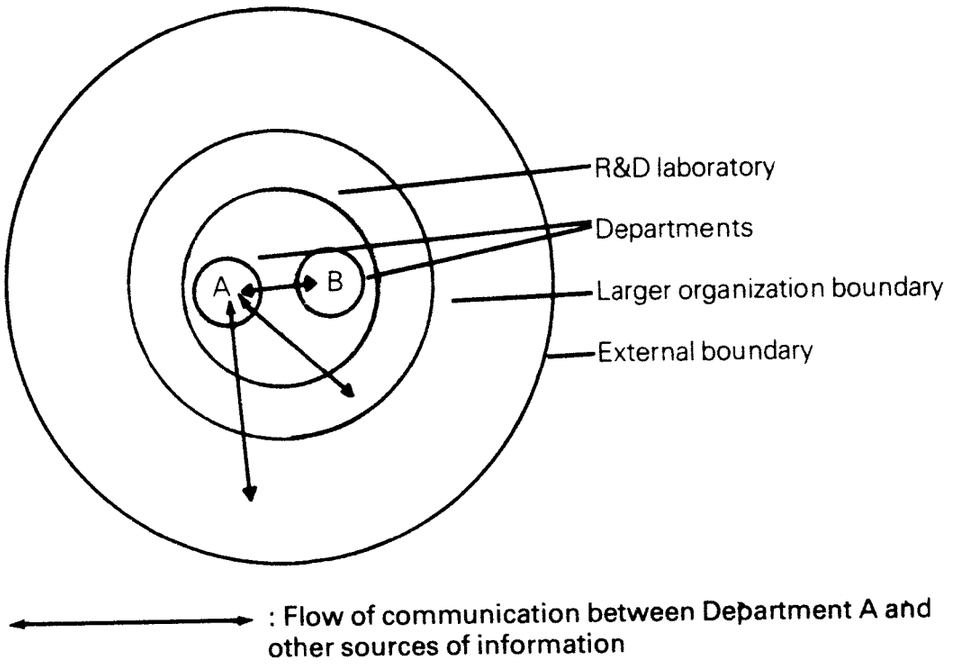

Figure 2. Communication boundaries in R&D laboratories.

information to several external information domains. Figure 2 distinguishes three external sources of information and thus three communication boundaries for work areas within an R&D laboratory — the extra-organization boundary, the laboratory to larger organization boundary, and the intra-laboratory boundary. While these boundaries are specifically relevant to R&D settings, the importance of transferring information across analogous organizational boundaries exists for all innovating systems (Baldridge and Burnham, 1975; Czepiel, 1975; Van de Ven, Delbecq, and Koenig, 1976). The following section discusses the relevance of these three boundaries in the innovation process.

Extra-Organization Boundary

The literature on innovation in R&D organizations emphasizes the importance of extra-organizational communication since the laboratory must receive up-to-date information about market and technological developments. Among others, Schmookler (1966), Myers and Marquis (1969), and Von Hippel (1976a, 1976b) have stressed the importance of gathering information on user or market need, while Allen (1966, 1970), Achilladeles, Jervis, and Robertson (1971), and Utterback (1971) have emphasized gathering information on new technological products, processes, and knowledge.

Innovation-related research not focusing on the R&D organization also highlights the importance of communicating with information areas external to the innovating system (Miller, 1971; Keegan, 1974; Baldridge and Burnham, 1975; Czepiel, 1975). Moreover, the literature on the task environment suggests that a changing environment, often encountered by innovating systems, increases the need for effective communication with the organization's environment (Burns and Stalker, 1966; Lawrence and Lorsch, 1967; Aiken and Hage, 1971; and Roberts *et al.,* 1974).

Laboratory to Larger Organization

The importance of this communication boundary is twofold. If market or technical information is important in the innovation process, and if marketing, sales, or manufacturing

have relevant information, then the successful projects will benefit from an effective laboratory to larger organization interface (Mansfield and Wagner, 1975; Rothberg, 1976). Further, as the project moves through the laboratory, it can benefit from technical input, support, and feedback from several organizational areas.

Innovation-related research not focusing on the R&D laboratory also supports the importance of intra-organizational coordination (Lawrence and Lorsch, 1967; Hage, Aiken, and Marrett, 1971; Hage, 1974). Further, to the extent that innovation builds in reciprocal task interdependence between subunits, the need for effectively integrating these subunits will be substantial (Thompson, 1967; Galbraith, 1973).

Intra-Laboratory Boundary

Frequently the innovating subunit is itself highly differentiated (Rosenbloom and Wolek, 1970; Lorsch and Morse, 1975). Substantial research exists demonstrating the importance of effectively transferring information within differentiated R&D laboratories. Utterback (1971) found that while sources outside the laboratory were most useful in generating new ideas, sources inside the laboratory were most helpful in providing problem solving and implementation assistance. Similarly, research by Allen (1966, 1970), Pelz, and Andrews (1966), Smith (1970), Farris (1972), and Goldhar, Bragaw, and Schwartz (1967) supports the importance of intra-laboratory communication links to take advantage of in-house technical expertise.

In all, this review suggests that to develop successful innovations, the innovating unit must attend to information and feedback from several external information domains. Much of the literature has concentrated on the extra-organizational communication boundary (Utterback, 1974). This review suggests, however, that work areas must attend to communication boundaries within the organizational as well (see Figure 2).

COMMUNICATION ACROSS ORGANIZATIONAL BOUNDARIES AS A TWO-STEP PROCESS

While communication across organizational boundaries is vital to each phase of the innovation process, much literature emphasizes the difficulties, inefficiency, and hazards of transferring information across organizational interfaces (Dearborn and Simon, 1958; Wilensky, 1967; Allen and Cohen, 1969; Roberts *et al.*, 1974). Why should these difficulties exist?

Systems theory suggests that as organizations grow, they differentiate. Specialized areas evolve to deal with particular tasks or environments (Katz and Kahn, 1966; Thompson, 1967). As these specialized units develop, each generates its own idiosyncratic norms, values, time frame, and coding schemes to permit effective processing of information (March and Simon, 1958; Katz and Kahn, 1966). Thus, not only will different organizations have a mismatch in coding schemes, but differentiated subunits within the same organization will likely also have contrasting languages and coding schemes.

These inherent conceptual and linguistic differences act as a communication impedance or as a communication boundary hindering the free flow of information. The greater the differentiation, the greater the communication impedance Rogers and Shoemaker, 1971; Miles, 1976). Supporting this communication mismatch logic, studies by Allen (1964, 1966), Schilling and Bernard (1964), and Baker, Siegmann, and Rubenstein (1967) found an inverse relationship between the amount of an individual's extra-organizational communication and his or her work performance.

If communication across organizational boundaries is difficult and prone to bias and distortion, yet is simultaneously a requirement for successful innovation, then how are innovating subunits effectively linked to external information areas? One way to deal with the difficulties of communicating across organizational boundaries is to develop special boundary roles (March and Simon, 1958; Thompson, 1967; Allen and Cohen, 1969; Allen, 1970; Aldrich and Herker, 1977; Schwartz and Jacobson, 1977). Individuals filling these roles are capable of translating contrasting coding schemes and therefore of acting as boundary spanners between the work unit and external information areas. Thus, information may flow into the R&D laboratory in a two-step fashion, through a set of key persons who channel this information to their colleagues within the laboratory.

This two-step flow of information across organizational boundaries is analogous to sociological research dealing with social influence, voting behavior, and the diffusion of innovation (Lazarsfeld, Berelson, and Gaudet, 1948; Coleman, Katz, and Menzel, 1966; and Rogers and Shoemaker, 1971). This research found evidence of opinion leaders, who because of their characteristics were well connected to external sources of information. These opinion leaders were sought after for advice and, in turn, influenced the decisions of the less active members of the social system.

These converging results suggest that opinion leaders or those filling special boundary roles, evolve to mediate the transfer of information from external information areas into the organization's internal communication network. While the idea of boundary roles is not new, much of the literature has focused on the extra-organizational communication boundary (Allen and Cohen, 1969; Frost and Whitley, 1971; Keller and Holland, 1975a, 1975b). This paper contributes to the boundary-role literature by extending the gatekeeper research, as well as by investigating the existence of boundary-spanning roles attending to other communication domains (see Figure 2). This paper then investigates the distribution of these boundary roles within the R&D laboratory. Specific hypotheses appear below.

Existence of Special Boundary Roles

Gatekeepers. Based on the social influence studies of Lazarsfeld, Berelson, and Gaudet (1948), Allen (1966) and Allen and Cohen (1969) suggested that key nodes in the innovating system's communication network convey information from external domains into the internal communication network. Allen and Cohen (1969) identified internal communication stars[2] and found that they were more

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Internal communication stars (the term is borrowed from the early sociometric research) are those individuals most frequently approached as technical discussion partners. See methods section for greater detail.

strongly connected to external sources of information than nonstars. These internal communication stars both were consulted more frequently *and* were well connected to external information areas. Those internal stars who also had a substantial amount of extra-organizational communication were termed gatekeepers. Allen (1970) replicated these findings in a large aerospace firm.

While the gatekeeper idea has stimulated extensive research (Pettigrew, 1972; Whitley and Frost, 1973; Taylor and Utterback, 1975), a question can be raised as to the existence of specializtion among gatekeepers: Do all gatekeepers attend to all external areas, or is there specialization by information domain (Allen and Cohen, 1969; Kelly and Kranzberg, 1975)? It may be that due to task requirements and different information needs, gatekeepers in different areas tend specific information domains (Allen and Cohen, 1969; Walsh and Baker, 1972; Whitley and Frost, 1973).

This idea of specialized gatekeepers can be made more exact. Since research projects require state-of-the art technical input, they must be well connected to professional areas outside the laboratory (Myers and Marquis, 1969; Utterback, 1971, 1974). Given the importance of this professional link, it may be that research gatekeepers direct their external communication specifically to universities, professional societies, and to the professional literature. On the other hand, technical service areas require up-to-date information on new products, processes, and markets (Achilladelles, Jervis, and Robertson, 1971; Czepiel, 1975). If so, gatekeepers in the technical service areas may focus their external communication on suppliers, vendors, and customers. This logic suggests that gatekeepers may not attend to all external communication areas, but may specialize in those external areas most critical to the work of their subunit.

Due to the location of relevant information, one can reason that for subunits engaged in research the professional communication boundary (universities or professional societies) is more critical than the operational communication boundary (suppliers, vendors, customers). The reverse should hold true for the technical service areas. If so, it can be hypothesized that:

Hypothesis 1. Gatekeeper specialization will occur by task area. Stars in research areas will have significantly more communication than nonstars with professional areas outside the organization. Similarly, stars in technical service areas will have significantly more communication than nonstars with operational areas outside the organization.

Organizational liaisons. Studies on the diffusion of innovation (Langrish, Gibbons, and Evans, 1972; Czepiel, 1975) as well as on the process of innovation (Utterback, 1971; Baldridge and Burnham, 1975; Von Hippel, 1976) emphasize the importance of the boundary between the innovating subsystem and the larger organization. In R&D settings, the communication boundary between the laboratory and the marketing, manufacturing, and sales areas is critical (Gruber, Poengsen, and Prakke, 1973; Mansfield and Wagner, 1975).

As discussed earlier, task and coding differences make communication across organizational boundaries difficult.

Given this inherent communication impedance, it may be that special boundary roles evolve to mediate communication between the laboratory's internal network and the larger organization. These intra-organizational boundary roles may perform a function similar to that of the integrator in Lawrence and Lorsch's (1967) work; boundary roles are different only in that they operate at a lower level in the organization and are not formally defined. Similarly, what Myers and Marquis (1969) term key people or product champions are probably more than particularly active individuals; they may act as key nodes in the information network, bridging the laboratory to other organizational areas. Using the two-step flow logic, one can hypothesize that communication between the R&D laboratory and the larger organization will not be direct but will be mediated by organizational liaisons.

Hypothesis 2. Internal communication stars will have significantly more communication with areas in the larger organization than nonstars.

Laboratory liaisons. Innovating subsystems may themselves be highly differentiated. If there is task interdependence between subsystems, then these units face the problem of internal integration (Lorsch, 1965). If R&D organizations are differentiated, they face an accentuated problem of integration given both task interdependence and the need for feedback from different laboratory areas (Pelz and Andrews, 1966).

With the same logic used for gatekeepers and organizational liaisons, one can hypothesize that because of the communication boundary between differentiated task areas, communication across even intra-laboratory interfaces occurs in a two-step fashion. Boundary roles may connect the department to other areas of the laboratory. This reasoning suggests that communication between differentiated task areas in the laboratory is not equally distributed among the technical staff but occurs through a few key network nodes — laboratory liaisons.

Hypothesis 3. Internal communication stars will have significantly more communication with other areas, that is, departments, of the laboratory than nonstars.

Note that to test Hypotheses 1, 2, and 3, communication stars as a set will be contrasted with nonstars. If significant differences between the two groups are found, that is evidence for the two-step flow of communication.

Distribution of Special Boundary Roles

The preceding discussion has hypothesized the existence of several kinds of boundary roles. What, however, is the distribution of these roles within the innovating system? The final set of hypotheses deals with this question.

As discussed earlier, boundary roles evolve to deal with the difficulties of gathering and diffusing information across organizational boundaries. However, increased work-related uncertainty may accentuate the difficulties of cross-boundary communication; the need for extra-unit communication will increase (Lawrence and Lorsch, 1967), while the ability to deal with the communication impedance may well decrease given the more complex problem-solving require-

ments in the work area (Lorsch and Morse, 1975; Miles and Perreault, 1976). If increased task uncertainty accentuates both the need for extra-unit information as well as the difficulties of extra-unit communication, it can be hypothesized that under such conditions, the number of boundary roles will increase to deal with the greater information-processing requirements (Keller and Holland, 1975a; Aldrich and Herker, 1977).

This reasoning implies that special boundary roles not only function as links to external sources of information, but that they also buffer the subunit from external uncertainty (Thompson, 1967). If so, the greater the subunit's work-related uncertainty, the greater will be the number of roles to deal with that uncertainty. Task characteristics, task environment, and task interdependence are each sources of work-related uncertainty (Pennings, 1975; Van de Ven, Delbecq, and Koenig, 1976). The distribution of special boundary roles, then, will not be equal throughout the innovating system, but will be contingent on the nature of the system's work. Lastly, if this logic has merit, work areas which fit the number of boundary roles to the demands of their work will be high performers (Galbraith, 1973; Tushman and Nadler, forthcoming).

This core hypothesis leads to the following more specific hypotheses:

Hypothesis 4a. Research projects will have more boundary roles per project than technical service projects.

Hypothesis 4b. Projects facing a changing environment will have more roles per project than projects facing a stable environment.

Hypothesis 4c. Projects with substantial task interdependence will have more roles per project than projects with little task interdependence.

SETTING AND METHODS

This study focused on the communication activity of all the professionals ($N=345$) in the R&D facility of a large midwestern U.S. corporation. The laboratory is isolated from the rest of the corporation and is divided into seven departments (divisional laboratories). The departments are, in turn, organized into separate project areas. The projects were stable over the course of the research and each respondent was a member of only one project.

Technical Communication

To provide communication data, each professional was asked to report his or her actual, work-related, oral communication for a number of selected days (the respondents were asked not to report strictly social communication, nor did they report written communication). These data were collected once a week for 15 weeks, with an equal number of each of the weekdays chosen. The overall response rate was 93 percent; moreover, 68 percent of all the communication reported within the laboratory was mentioned by both parties (cf. Weiss and Jacobson, 1960). The extra-laboratory data were self-report data which could not be corroborated with discussion partners.

In order to identify the internal communication stars and to test for boundary roles, the communication activity of each

professional was separated into the following mutually exclusive communication domains — intra-department, intra-laboratory (outside the department), organizational, and extra-organizational. Extra-organizational communication was further separated into a professional domain (universities, professional societies) and an operational domain (suppliers, vendors, customers). The literature has used a number of criteria to define internal communication stars (Allen, 1970; Whitley and Frost, 1973). This study arbitrarily defined internal stars as those individuals in the top fifth of the intra-department communication distribution. Since one of the departments was substantially larger than the other six, it was excluded from these analyses. Fifty stars were located by these methods.

Project Task Characteristics

A number of analysts have suggested that task characteristics can be ordered on a dimension from complex (non-routine) to simple (routine). In R&D settings, tasks can differ by (1) length of time required for feedback, (2) specific problem versus general problem orientation, and (3) their generation of new knowledge in contrast to their use of existing knowledge (Rosenbloom and Wolek, 1970). With these attributes, four task categories were developed with the laboratory's managers to form a task dimension ranging from complex (research) to less complex (technical service) (see Table 1).

Table 1

Categories of Task Characteristics

I. Basic Research	II. Applied Research	III. Development	IV. Technical Services
Work of a general nature intended to apply to a broad range of applications or to the development of new knowledge about an area.	Work involving basic knowledge for the solution of a particular problem. The creation and evaluation of new concepts or components but not development for operational use.	The combination of existing or feasible concepts, perhaps with new knowledge, to provide a distinctly new product or process. The application of known facts and theory to solve a particular problem through exploratory study, design and testing of new components or systems.	Cost/performance improvement to existing products, processes or systems. Recombination, modification and testing of systems using existing knowledge. Opening new markets for existing products.

Using these categories, respondents answered two questions about their work:

(1) In which of these categories (see Table 1) would you place the objectives of your work? The Arabic numbers indicate the degree of refinement within each category, where 1 is "pure."

I	II	III	IV
1 2 3	1 2 3	1 2 3	1 2 3

(2) Roughly what percentage of the work required to accomplish the task's objectives falls in each of the four categories?

I	II	III	IV

The individual responses to the two questions were correlated highly (Spearman-Brown reliability=.91). Thus, individual task scores were calculated by taking the average of the two questions. Project scores were calculated by pooling individual scores; 58 projects provided usable responses.

Project Task Environment

Of the many environmental dimensions studied (cf. Jurkovitch, 1974), Duncan (1972) and Negandi and Reimann (1973) suggest that the dimension showing the range between stability and change is a particularly important contributor to perceived uncertainty. Following this research, only the stable-changing dimension of the task environment was investigated. Each respondent answered the following question:

We are interested in how rapidly you see the demands of your job changing. To what extent are techniques or skills or information needed for your project(s) changing?

1	2	3	4	5
To a very little extent	Little	Some	Great	To a very great extent

Fifty-eight projects provided usable responses. Again, project scores were calculated by pooling individual scores.

Project-Task Interdependence

Project-task interdependence refers to the extent to which a task requires working with other areas of the organization. If the focal organization is itself highly differentiated, then it is necessary to distinguish between the amount and the location of the interdependence. The amount and location of task interdependence were operationalized in the following fashion:

On the scales below, please indicate the degree to which your work is interrelated with the work of:

1. technical personnel in *other departments.*

1	5	10
very low interrelationship	moderate	very high interrelationship

2. technical personnel in the organization but *outside of the laboratory.*

1	5	10
very low interrelationship	moderate	very high interrelationship

Fifty-eight projects provided usable responses. Project scores were calculated by pooling individual scores.

Unit of Analysis

Since the unit of analysis in Hypothesis 4 is the project, individual responses must be pooled to get project scores. To check the appropriateness of pooling, the homogeneity of project members' perceptions of each variable was tested. A one-way analysis of variance was done to detect any significant differences among individual perceptions; Bartlett's M-test (Dixon and Massey, 1969) was used to test specifically for the homogeneity of intra-project perceptions. For variables which passed both of the above tests, a more stringent test was performed on each project. Each project's variance was compared to the pooled variance for the particular variable studied. If a project's variance was significantly greater ($p<.01$) than the pooled variance, then that project was not used in the analysis of that particular variable.

The ANOVA and Bartlett tests indicated that pooling was appropriate for each of the variables. The individual F-tests

eliminated between two and four projects per variable. Project task scores fell into three distinct areas: research ($N=13$), development ($N=22$), and technical service ($N=20$).

Project Performance

Since performance measures are particularly difficult to develop for R&D settings (Whitley and Frost, 1971), a subjective measure similar to that used by Lawrence and Lorsch (1967) was employed. All group managers ($N=7$) and both laboratory directors ($N=2$) were interviewed individually. Each was asked to evaluate the overall technical performance of all projects with which he was technically familiar. Since each project was developing new technologies or new products or revising existing products, this performance measure can be taken as a proxy for the innovativeness of the project. An average of 4.7 managers independently rated each project on a seven-point scale. Since the intercorrelations of the nine judges correlated moderately (Spearman-Brown reliability=.81), individual ratings were averaged to yield overall project scores.

RESULTS

Existence of Special Boundary Roles

To test for the existence of gatekeepers (Hypothesis 1), organizational liaisons (Hypothesis 2), and laboratory liaisons (Hypothesis 3), the amount of external communication of stars and nonstars must be compared. Table 2 compares the median absolute amount of communication, over 15 weeks, of stars and nonstars with the three extra-departmental communication domains.[3] The data are organized by department. There is strong support for the existence of both the organizational and laboratory liaisons: 11 of the 12 differences between stars and nonstars are statistically significant. Stars, as a set, have significantly more communication than the nonstars with the larger organization as well as with other areas in the laboratory. Therefore, as predicted, communication between organizationally distant areas is not direct but occurs in a two-step fashion; that is, the majority of the technical staff keep informed of extra-departmental issues by speaking with the internal stars who, in turn, are strongly linked to other areas in the larger organization.

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Because of the skewed distributions, Mann-Whitney U-tests are used. Since each of the hypotheses predicts the direction of difference, one-tailed tests are utilized. (As the numbers of cases in the larger groups are all greater than 20, the power-efficiency of these tests approaches that of equivalent t-tests [Siegel, 1956]).

Table 2

Median Amount of External Communication by Stars (st) and Nonstars (nst)†

Communication between:	DEPARTMENTS											
	Control Systems		Special Products		Product Analysis		Construction Development		Development		Research	
	st $N=(10)$	nst (43)	st (6)	nst (22)	st (7)	nst (29)	st (7)	nst (28)	st (11)	nst (38)	st (9)	nst (33)
Laboratory	4	0••	22	18	37	14•••	20	6•••	12	3•••	23	12••
Organization	19	6•••	26	10••	23	7••	37	21•••	3	1••	3	1•
Extra-organization	7	1••	3	3	4	2	7	7	4	1••	3	1

Mann-Whitney U-Test — one tail:

• $p<.10$

•• $p<.05$

••• $p<.01$

†Tables 2 and 3 report the median absolute amount of external communication for stars and nonstars over a 15-week period.

While support for both organizational and laboratory liaisons is strong, support is weak for the gatekeeper role (see Table 2). In only two departments do the stars have significantly more communication than the nonstars with areas outside the organization. However, Hypothesis 1 argued that gatekeepers specialize in specific external information domains. It was hypothesized that given the need in research areas for state-of-the-art technical information, research stars would focus their extra-organizational communication on the professional domain (universities, professional societies). On the other hand, it was hypothesized that technical service stars would focus on operational areas (suppliers, vendors, and customers) given the technical service areas' need for up-to-date information on new products and processes.

Table 3

Median Amount of Extra-organizational Communication by Stars (st) and Nonstars (nst)

	Technical Service Department		Research Departments			
Communication between:	Control Systems st	Control Systems nst	Development st	Development nst	Research st	Research nst
Operational areas:	6	1••	2	1	1	1
Professional areas	0	0	2	0••	4	0•

Mann-Whitney U-Test — one tail:

• $p<.10$

•• $p<.05$

To test this specialization hypothesis, research departments and technical service departments had to be located, and extra-organizational communication separated into professional and operational categories. In this laboratory, the Research and the Development Departments were made up solely of research projects, while only the Control Systems Department was composed solely of technical service projects.[4]

For these three departments then, Table 3 compares the median absolute amount of communication to operational and professional areas for stars and nonstars. The data indicate support for the gatekeeper specialization hypothesis. In the two research departments, the stars have significantly more communication to professional areas than the nonstars; there is no role specialization to the operational domain for these research departments. In the technical service department, however, stars have significantly more communication than nonstars to the operational domain. And, as hypothesized, there is no role specialization to the professional domain for this technical service department.

These analyses, therefore, support the existence of several boundary-spanning roles. Persons filling these roles serve an important function in the innovation process in that they link their work areas to several external information domains. The laboratory and the organizational liaisons span

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Various kinds of projects made up the other departments. Since task heterogeneity would confound the analysis, these other departments could not be used to test the gatekeeper-specialization hypothesis.

Table 4

Mean Number of Boundary Roles per Project for Different Task Areas•

	Research	Technical Service	Significance	Degree of Freedom
Total	.64(.67)	.50(.56)		(19)
Performance:				
High	.80(.84)	.50(.84)		(9)
Low	.50(.71)	.50(.71)		(8)

•
For Tables 4, 5, and 6 boundary roles were isolated by taking those internal communication stars which were also connected strongly to an external information domain (top fifth of the external communication distribution). Standard deviations are within parentheses.

Table 5

Mean Number of Boundary Roles per Project for Different Environmental Conditions

	Environment Changing	Environment Stable	Significance	Degree of Freedom
Total	1.00(.81)	.39(.60)	••	(32)
Performance:				
High	.86(.69)	.00(.00)	•••	(12)
Low	1.11(.93)	.64(.67)		(18)

t-tests:
•• $p<.05$
••• $p<.01$
(Standard deviations are within parentheses.)

Table 6

Mean Number of Boundary Roles per Project for Different Amounts of Task Interdependence

	Interdependence‡ High	Interdependence‡ Low	Significance	Degree of Freedom
Total	.94(.70)	.50(.77)	•	(32)
Performance:				
High	.83(.82)	.12(.73)	•	(12)
Low	1.10(.95)	.70(.46)		(18)

t-tests:
• $p<.10$
‡Interdependence is the sum of laboratory and organizational interdependence.
(Standard deviations are within parentheses.)

intra-organizational boundaries, while the gatekeepers span the extra-organizational boundary. However, because of the differing information needs of the various task areas, gatekeepers do seem to specialize in particular external information domains. Research gatekeepers focus on professional areas, while technical service gatekeepers focus on operational areas.

Distribution of Special Boundary Roles

These results indicate the existence of special boundary role

holders who attend to several communication boundaries. What, however, is the distribution of these roles within the laboratory? Based on information processing ideas, Hypothesis 4 suggested that the greater the subunit's work-related uncertainty, the greater the number of special boundary roles to deal with that uncertainty.

Tables 4, 5, and 6 support this core hypothesis. Projects with more complex information-processing requirements consistently have more boundary roles than projects with less complex information processing requirements. In the total sample, research projects have more boundary-spanning roles than technical service projects; projects facing a changing environment have significantly more roles per project than projects facing a stable environment; and projects with substantial task interdependence have significantly more roles per project than projects with little task interdependence. Evidently, under routine technical conditions, a small number of boundary roles can meet the external information requirements, while more complex or uncertain technical conditions require more boundary roles to attend to the increased uncertainty. Thus, boundary roles do seem to be a way of dealing with work-related uncertainty and are, therefore, distributed differentially within the laboratory, with distribution being contingent on the nature of the project's work.

But what is the relationship between the number of boundary roles per project and project performance? If the logic of Hypothesis 4 holds, then the results found for the total sample should be accentuated in high-performing projects. Tables 4, 5, and 6 contrast the number of roles per project for high- and low-performing subsets (the performance scores were split at the median). As predicted, the results are accentuated for high-performing projects: high-performing projects facing greater information processing requirements (projects performing research tasks, facing a changing environment, or substantial interdependence) have more boundary roles per project than high-performing projects facing less uncertainty. However, for the low-performing projects, the hypothesized differences decrease in magnitude and do not approach statistical significance.

The patterns of Tables 5 and 6 suggest that there can be an excess of boundary roles even under conditions of substantial work-related uncertainty. High-performing projects facing a changing environment or substantial task interdependence have fewer roles per project than low-performing projects facing the same task conditions. Evidently, high-performing projects facing substantial work-related uncertainty tend to share their boundary roles. In contrast, low-performing projects facing a changing environment and/or substantial task interdependence tend to have more than one boundary role per project. It may well be that having a boundary role for each project is redundant and associated with problems of coordination and integration.

These ideas suggest a curvilinear relationship between the number of boundary roles and project performance for projects facing substantial work-related uncertainty (due either to the task environment or the demands of task interdependence). Supporting this idea, Figure 3 shows that for

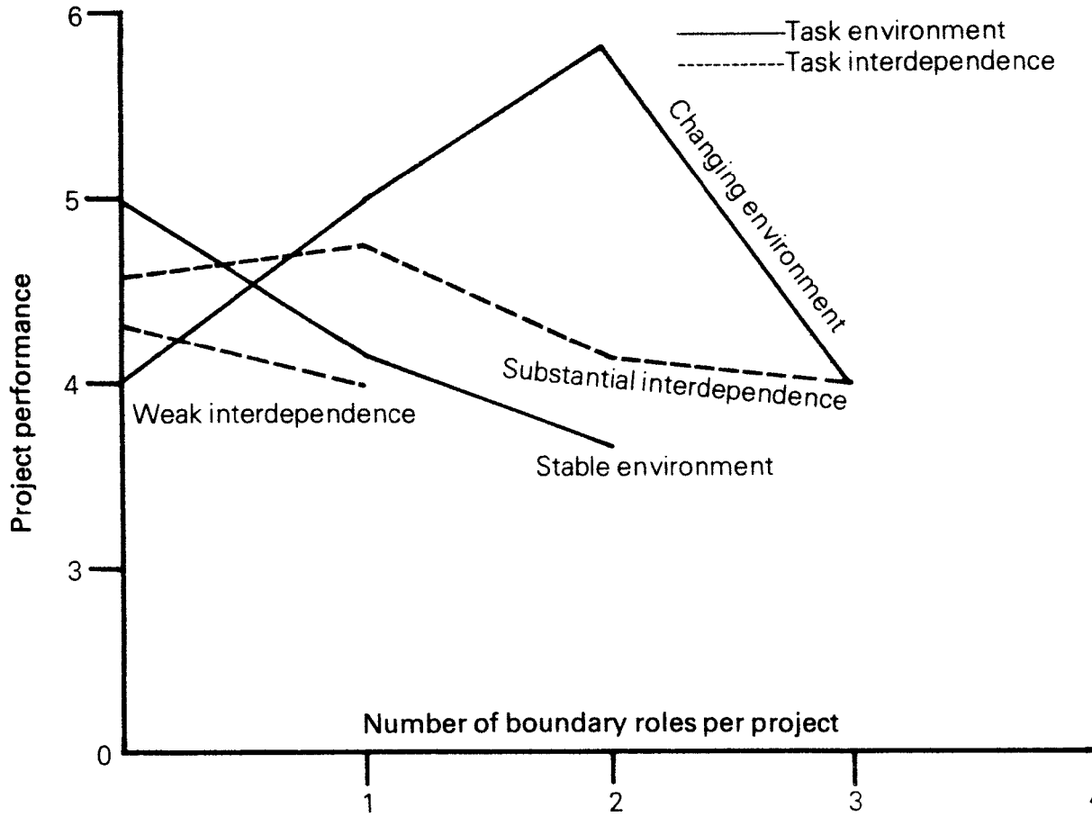

Figure 3. Average project performance for different numbers of boundary roles per project (*N* = 16).

such projects, an increase in the number of boundary roles per project is associated initially with an increase in project performance. After a point, however, an increased number of boundary roles is associated with decreased project performance. In further support of the ideas behind Hypothesis 4, Figure 3 also shows that projects facing smaller information processing requirements have fewer boundary roles. For these projects, an increase in the number of boundary roles is consistently associated with lower project performance. Thus, under conditions of both minimal uncertainty and substantial uncertainty, too many boundary roles may be redundant and actually impede coordination and control. As Figure 3 is made up of a limited number of points, these results should be seen as suggestive of a curvilinear relationship between the number of boundary roles and project performance, and as a stimulus for future research.

In review, boundary roles evolve to deal with several communication boundaries. These analyses suggest that the number of boundary roles per work area is dependent on the characteristics of the area's work. More roles per project are not always related to higher performance. For high-performing projects, the number of boundary roles per project is contingent on the nature of the project's work. The more the uncertainty in the task, the greater the information-processing requirements and, up to a point, more roles evolve to deal with that increased uncertainty. Thus, beyond the cross-boundary communication function, boundary roles also seem to be an effective way of dealing with extra-unit uncertainty. Finally, the results indicate that work characteristics do not cause boundary roles; rather, the high-performing work areas develop the appropriate number of boundary roles to fit the external information-processing demands of their work.

CONCLUSIONS AND IMPLICATIONS

This research is based on the premise that the process of innovation is sensitive to the effective transfer of information across organizational boundaries. Yet because of the consequences of differentiation, widespread communication across boundaries tends to be inefficient and prone to distortion. This study has investigated one mechanism for efficiently linking an innovating subsystem, an R&D laboratory, to its diverse sources of external information.

This research indicates that special boundary roles function to link the innovating system with various sources of external information and feedback. Thus, communication with external areas is not distributed equally among the innovating unit's staff but takes place through a limited set of individuals able to translate between several coding schemes. These individuals, or special boundary roles, are well connected to external information areas and are frequently consulted within the innovating unit. These boundary roles exist to mediate communication across several organizational interfaces.

These results support the idea that information from external areas enters innovating units in an indirect, two-step fashion. Similarly, this research supports the organizational work on boundary-spanning roles (March and Simon, 1958; Thompson, 1967; Galbraith, 1973; Miles and Perreault, 1976; Aldrich and Herker, 1977; Schwartz and Jacobson, 1977). Finally, this research indicates that while too many boundary roles may be inefficient, for high-performing projects, the number of special boundary roles is nevertheless contingent on the nature of the innovating unit's work.

These results suggest several directions for future research. Most obviously, future research can develop a stronger operationalization of the boundary-spanning role. Ideally, one would like a multidimensional measure which taps formal as well as informal aspects of the role. Future research can also focus on the characteristics of these boundary roles and what functions they actually perform. Do persons filling these roles simply transfer (or filter) information, or are they involved in a wider range of activities, given their exposure to scarce information (Pettigrew, 1972; Kelly and Kranzberg, 1975). Other research questions concern the coordination of several boundary-spanning roles and developing these special roles over a period of time. Lastly, while the literature relating individual differences to boundary-role status has not been fruitful (Farris, 1972; Keller and Holland, 1975b; Allen, 1977), research by Miles and Perreault (1976) and Keller, Szylagyi, and Holland (forthcoming) suggests that an interesting avenue of research is the relation among boundary-role status, individual differences, and role conflict.

From a more pragmatic point of view, these results have several implications for managers. Most generally, those interested in managing innovation should explicitly recognize the importance of key individuals in the system's communication network. Not only are special boundary roles an efficient mechanism for gathering and transmitting information, but evidence also suggests that they are important in

other organizational areas (for example, as integrators or as implementors of management information systems), given their exposure to critical information and their informal status (Rogers and Shoemaker, 1971; Pettigrew, 1972; Kelly and Kranzberg, 1975; Schwartz and Jacobson, 1977).

Beyond recognizing the existence and the importance of special boundary roles in the innovation process, this research suggests that managers can actively encourage the development of boundary roles. Recognizing and rewarding boundary-spanning activity are most important, particularly if the role is associated with stress and conflict (Miles and Perreault, 1976). In terms of facilitating boundary-role status, managers can ease the access of their high perfomers to both areas of external information (for example, marketing, professional meetings) and to professional literature. Similarly, managers can facilitate extensive communication networks for their high-performing individuals through a set of job assignments (Edstrom and Galbraith, 1977).

Lastly, this research suggests that managers be sensitive to the impact of task characteristics on boundary roles. The number of boundary roles per work area should be tailored to meet the unit's information processing needs, and different task areas may require boundary roles with particular backgrounds and characteristics (Whitley and Frost, 1973). It is not clear that promotion or formal boundary-role status facilitates boundary-role activities (Allen, 1970). However, these several implications do suggest that managers can enhance the process of innovation by recognizing the importance of boundary roles and working to actively develop and use their organization's emergent or informal communication systems.

In conclusion, given their ability to scan and interpret information from external areas and to transfer this information to the innovating unit, persons filling boundary roles can be seen as an important information processing mechanism in the innovation process. Of course, this research is limited by the type of organization studied. However, if the underlying logic has merit, similar results should be found in non-R&D settings. Future research must extend and replicate these results as well as deal with other organizational processes which promote the flow of information so critical to the innovation process.

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